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ORIGINAL ARTICLE

LIMFA: label-irrelevant multi-domain feature alignment-based fake news detection for unseen domain

Danke Wu[1] · Zhenhua Tan[1] · Haoran Zhao[1] · Taotao Jiang[1] · Meilin Qi[1]

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Abstract

Fake news in social networks causes disastrous effects on the real world yet effectively detecting newly emerged fake news remains difficult. This problem is particularly pronounced when the testing samples (target domain) are derived from different topics, events, platforms or time periods from the training dataset (source domains). Though efforts have focused on learning domain-invariant features (DIF) across multiple source domains to transfer universal knowledge from the source to the target domain, they ignore the complexity that arises when the number of source domains increases, resulting in unreliable DIF. In this paper, we first point out two challenges faced by learning DIF for fake news detection, (1) high intra-domain correlations, caused by the similarity of news samples within the same domain but different categories can be higher than that in different domains but the same categories, and (2) complex inter-domain correlations, stemming from that news samples in different domains are semantically related. To tackle these challenges, we propose two modules, center-aware feature alignment and likelihood gain-based feature disentanglement, to enhance the multiple domains alignment while enforcing two categories separated and disentangle the domain-specific features in an adversarial supervision manner. By combining these modules, we conduct a label-irrelevant multi-domain feature alignment (LIMFA) framework. Our experiments show that LIMFA can be deployed with various base models and it outperforms the state-of-the-art baselines in 4 cross-domain scenarios. Our source codes will be available upon the acceptance of this manuscript.

Keywords Fake news detection · Social media · Domain generalization · Feature disentanglement

1 Introduction

With the advancement in web technologies, social media has become the main platform for people to share information and acquire news. Unfortunately, few users verify the veracity of the content they share, making online social networks an ideal environment for spreading fake news. As reported in the *2022 Edelman Trust Barometer*,[1] the widespread spread of fake news has resulted in distrust of major information sources, with owned media earning 43% of trust compared to 37% for social media. And now, 76% of people are concerned that fake news or false information is being used as a weapon. Under such circumstances, automatic fake news detection is critical for rebuilding trust in social media and sustaining social stability.

Previous prevailing solutions have employed data-driven deep learning approaches to extract features from users, social context and external knowledge for fake news detection [4, 8, 14, 15, 17, 41]. Although these approaches yield satisfactory results during the training phase, they ignore the domain differences and therefore perform poorly when confronted with newly emerged news samples. In real-world applications, the testing dataset (target domain) and training dataset (source domain) are often derived from different time periods, platforms, topics and events. As a result, these datasets typically differ in data distributions, manifesting as subtle differences in word usage, diffusion structure, sentiment [25, 39], class distribution [34], etc. Previous models are easily trapped by domain differences

✉ Zhenhua Tan
tanzh@mail.neu.edu.cn

Danke Wu
wudk2019@stumail.neu.edu.cn

1 Software College, Northeastern University, Shenyang 110819, China

1 https://www.edelman.com/trust/2022-trust-barometer.

and make unreliable predictions. To address this problem, the domain adaptation approaches are employed to learn domain-invariant features (DIF, e.g. sensational language) shared between the source and target domains to form a universal decision boundary [2, 14, 19, 36]. Though generalization has improved, they necessitate training or fine-tuning with target domain samples, which introduces time delays and forfeits the opportunity to debunk fake news at its early diffusion stage. In contrast, domain generalization approaches strive to learn the DIF from multiple source domains to apply to any target domain [11]. In this study, we explore the domain generalization solution for cross-domain fake news detection, preparing the model for newly emerged news without using target samples.

Challenges In this direction, Castelo et al. [2] and Wang et al. [30] proposed EANN and BDANN, which employ a domain discriminator to establish feature alignment at the output level with the domain label engaged. The domain-specific features (DSF, e.g. specific vocabulary) are discarded, and only DIF are utilized for classification. Though they have been proven to be effective in aligning two domains, we argue that these studies overlook two critical challenges in aligning multiple domains. First, the ***high intra-domain correlations***. As sampling by topics illustrated in Fig. 1, a representative domain sampling, the news samples within Health domain exhibit strong semantic correlations, even surpassing the correlations between news samples from different domains but within the same category. Consequently, these high intra-domain similarities can lead to news samples from different categories being closely clustered in the embedding space (Fig. 1a), resulting in a fuzzy decision boundary. Second, the ***complex inter-domain correlations***. This issue arises because real-world domain categorization generally cannot comprehensively reflect the affiliation of news samples [39]. For instance, the society news involves finance (employees) and health (COVID-19) topics but is categorized into the society domain only. Such incomplete domain labels expose the model to incomplete feature alignment (Fig. 1b) and potential feature entanglement (Fig. 1c), causing sub-optimal performance.

Proposed approach To address these problems, we propose a center-aware feature alignment component to be label-irrelevant, which is inspired by contrastive learning [13, 34] that aligns the distribution of multiple domains at the feature level. To avoid costly pair data sampling, we adopt the strategy of learning category centers as anchors for feature alignment. The likelihood of the classifier is further introduced in feature alignment to expand the distance between different categories in the embedding subspace, alleviating the discrimination degradation of DIF during the alignment process. Additionally, we design a feature disentanglement component to facilitate the disentanglement between DSF and DIF by estimating the information gain of the DSF over the aligned DIF. Based on these two components, we constitute a label-irrelevant multi-domain feature alignment framework (denoted as LIMFA) for fake news detection. Experiments illustrate that our framework outperforms the state-of-the-art baseline on target domains without impairing its discrimination to source domains. Our main contributions are summarized as follows,

- We investigate the problem of learning generalizable features for fake news detection and identify two challenges, complex inter-domain correlations and high intra-domain correlations.
- To solve these challenges, we propose a label-irrelevant multi-domain feature alignment framework (LIMFA), which performs feature disentanglement and feature alignment simultaneously, improving performance to target domains while keeping discrimination on source domains.
- Experiment results demonstrated that our LIMFA improves the generalizability of a wide range of base models and surpasses the state-of-the-art baseline in

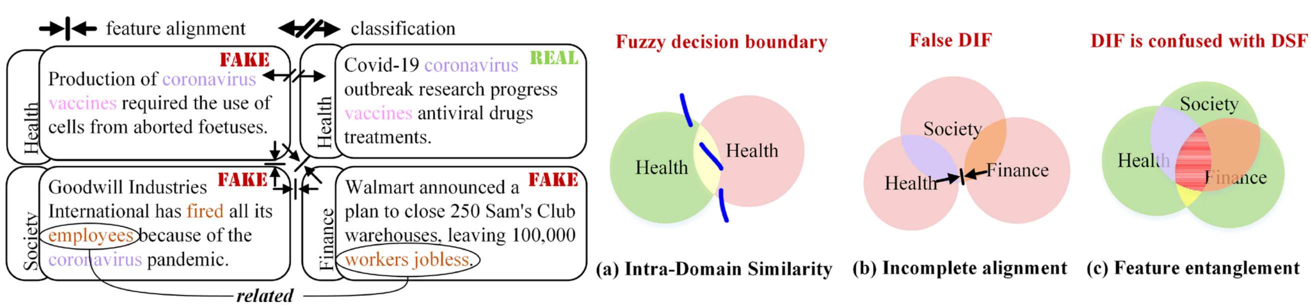


Fig. 1 An illustration of complex inter-domain and high intra-domain correlations. Owing to the complex correlations (words marked with the same color), the news sample labeled as Health news is indeed a mixture of Society and Health domains. Meanwhile, the semantic correlations within the Health domain are higher than that between the Health and Society domains. Ignoring such complex correlations, existing models suffer from (a) intra-domain similarity, (b)incomplete alignment and (c) feature entanglement and therefore fail to learn a clear and invariant decision boundary (Color figure online)

four fake news detection scenarios with 9 news domains (or 5 events). We further visualize the DIF under different models.

The structure of this article is organized as follows. Section 2 presents related works of experts in correlative areas. In Sect. 3, we describe our LIMFA. The results and visualizations of experiments are stated in Sect. 4, followed by the conclusion and future work.

2 Related work

In online social networks, fake news is an intentionally and verifiably false news piece and fake news detection aims to identify it from the real one [16]. We first delve into the literature that assumes an identical distribution between the training and testing datasets, which is referred to as single-domain fake news detection. In our context, the domain refers to a set of news samples governed by a specific sampling mechanism, such as a topic or an event, each with its unique distribution. Confronting with domain differences, multi-domain fake news detection aims to simultaneously modeling multiple domains within a model, while cross-domain fake news detection seeks to transfer general knowledge to target domains. We provide a summary of existing literature and outline our research objectives in Table 1. The following sections offer concise overviews.

2.1 Single-domain fake news detection

Earlier studies primarily concentrated on extracting informative features from data in various formats for a specific domain. The content of the texts or news images is most investigated. Various stylistic features were extracted at the syntactic, lexical, and semantic levels, such as total words, frequency of words and phrases, punctuation and parts-of-speech (POS) tagging [21, 24] for fake news detection. Furthermore, the semantic evidence likes linguistic styles [22, 26], novelty [10], and emotion [38] were then explored. In addition to text content, the social context of news pieces, including user profile [3, 33], user comment or repost [12], diffusion network [17], drew significant attention from researchers. Bian et al. [1] propose the BIGCN that incorporates the content semantics and bi-direction propagation clues of the comments. Lin et al. [12] proposed ClaHi-GAT, which enhances the representation of news samples by semantically inferring between the news claim and corresponding comments. The third kind of data explored is external knowledge, including the evidence collected from webpages or online encyclopedias that is relevant to the news piece [5, 20, 29] and the knowledge graph to complement the semantic representation of posts [6, 23, 35].

2.2 Multi-domain fake news detection

Some studies recognize that the real news stream encompasses news pieces from various domains, prompting the development of multi-domain fake news detection approaches to enhance overall performance. For instance, Nan et al. [19] propose MDFEND, which aggregates multi-expert networks that are dominated in different domains to improve the generalization of a certain model for all domains. Similarly, Zhu et al. [39] maintain a domain memory bank to model domain characteristics and enrich domain information which could discover potential domain labels based on seen news pieces. Silva et al. [25] propose a novel framework that jointly preserves domain-specific and cross-domain knowledge in news pieces to detect fake news from different domains. Despite their success in

Table 1 A summary of related literature and our research objective

Category	Objective	Methodology	Training			Testing
			Source domain	Domain label	Target domain data	Target Domain
Single-domain [12, 28, 35]	Improving performance	Neural network	A	–	–	A
Multi-domain [19, 25, 39]	Improving generalization for multiple source domains	Multi-expert system	A, B, C	✓	✓	A, B, C
Cross-domain [34, 36] (adaptation)	Improving generalization for a specific target domain	Adversarial learning Contrastive learning	$A, \bar{B}$	✓	✓	B
Cross-domain [30, 37] (generalization)	Improving generalization for any target domain	Adversarial learning	A, B, C	✓	–	$D, \ldots$
Ours	Improving generalization for source and target domains	Contrastive learning & Feature disentanglement	A, B, C	–	–	$A, B, C, D, \ldots$

improving the performance across all source domains, as new domains (events or topics) keep arising, their ability on newly emerged news pieces will diminish [42].

2.3 Cross-domain fake news detection

Cross-domain fake news detection seeks general knowledge from other domains to enhance the performance of the target domain. Two kinds of situations are considered for cross-domain fake news detection, one where limited samples of the target domain are accessable (known as domain adaption) and another where the target domain is totally unseen (known as domain generalization, ours). In the case of limited target data, some studies adopt domain adversarial learning to alleviate the domain differences[18, 23, 36]. For instance, Zhang et al. [36] proposed the MDDA model, which disentangles event-invariant style features from content features and adapts the style feature from the source domain to the target domain through a domain discriminator. An alternate solution adopts contrastive learning methods that minimize the distance between source and target domain samples of the same category while enlarging the distance between samples of different categories by measuring the cosine similarity between samples[13]. In particular, Yue et al. [34] leveraged the maximum mean discrepancy (MMD) distance to minimize the domain discrepancy to adapt misinformation systems to the COVID-19 domain. On the other hand, some studies retrieve DIF from multiple source domains rather than accessing the target domain. For instance, Wang et al. [30] proposed EANN, in which features are learned under the joint supervision of a fake news detector and a domain discriminator. As so EANN discards DSF and preserves DIF for classification Zhang et al. [37] then replaced the TextCNN in EANN with transformer and achieved the SOAT performance.

The differences between our work and existing studies are summarized in Table 1. With specific research objectives, these four categories of studies utilized different methodologies to train models on specialized training datasets and have demonstrated good performance in specific target domains. However, none of them have been able to effectively handle both the source and target domains. The studies [30] and [37], which are most similar to our studies, have made improvements in the generalization to any target domain. However, these approaches have limitations in terms of impairing discrimination on the source domains by ignoring DSF in the classification process[25]. Moreover, relying on incomplete domain labels hinders effective domain alignment, particularly when the number of source domains increases. To address these limitations, we introduce the feature selection gate to find two suitable subspaces for preserving DIF and DSF. Meanwhile, we develop a center-aware feature alignment module and a likelihood gain-based feature disentanglement module to enhance the alignment of multiple domains while separating two categories at the feature level and enforce adversarial feature disentanglement.

3 Proposed approach

3.1 Problem statement

In this paper, we define a special case of cross-domain fake news detection: Classifying the news pieces in the target domain D_{target} as fake or real while the target domain is unseen during the training phase. Given a multi-domain dataset $D_{soure} = \{x^n, y^n\}_n^N$, each news sample consists of its contextual data x and corresponding category label $y \in \{0, 1\}$, and N is the number of news samples. We assume that $f^k(x) = f^*(x) + \Delta^k$ is the model for the kth domain, where $f^*(x)$ can be regarded as a domain-invariant model with universal decision boundaries, and Δ^k is a domain-specific bias term. With no prior knowledge of the target domain D_{target}, we suppose to find a domain-invariant model $f^*(x) \rightarrow y$ from the multi-domain dataset, such that the $f^*(x)$ can be generalized to any target domain. In other words, the objective is to achieve a minimum prediction error in the target domain D_{target},

$$\min_{f^*} \mathbb{E}_{(x,y)\in D_{target}}[\mathcal{L}(f^*(x), y)] \tag{1}$$

where $\mathcal{L}$ is an appropriate loss function, $\mathbb{E}[\cdot]$ denotes the expectation of prediction error.

3.2 Framework overview

To achieve this, we propose a label-irrelevant multi-domain feature alignment framework (LIMFA). The overall architecture of LIMFA is shown in Fig. 2. It employs a gated feature disentanglement network as a backbone, including the feature extractor that learns the original features h_{all}, the feature selection gate that adaptively splits the original features into two parts(h_{ds}/h_{di}), and the restorer that restitutes data from features to prevent information loss. The proposed center-aware feature alignment ensures that h_{di} is invariant across all domains and separable between categories with the likelihood-based center loss $\mathcal{L}_{cc}$. The likelihood gain-based feature disentanglement enforces h_{ds} to be completely disentangled from h_{di} with the likelihood gain loss $\mathcal{L}_{lg}$. Two classifiers are responsible for making the DIF and DSF discriminative for fake news detection.

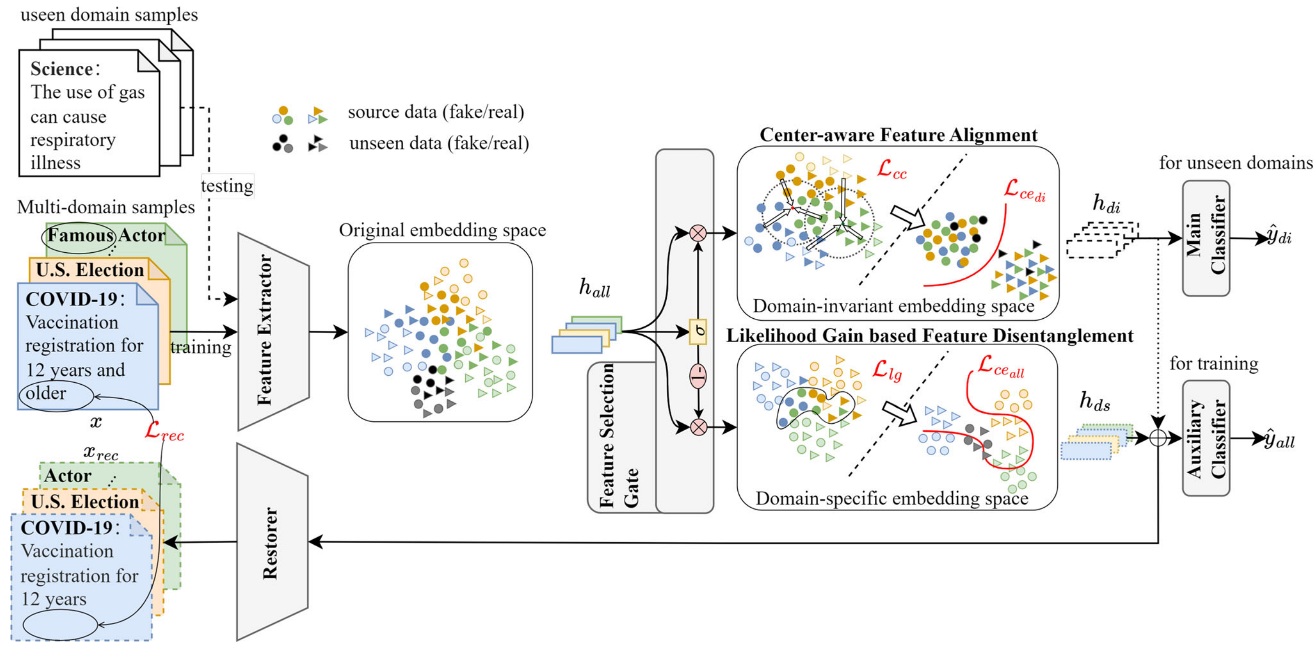


Fig. 2 Overview of the proposed LIMFA, in which the feature extractor learns overall features h_{all}, the feature selection gate splits the overall features h_{all} into the DIF h_{di} and the DSF h_{ds}, the restorer restitutes data from features to prevent information loss, the auxiliary classifier relied on DSF helps training, the main classifier relied on DIF is transferred to target domains. In the illustrated embedding spaces, each data point's shape and color denote its potential category and domain (light for DSF, and DIF otherwise), respectively

3.3 Backbone network

Feature extractor and feature selection gate In our LIMFA, each news piece is represented as input x. The feature extractor $\mathcal{E}$ aims at learning the discriminative feature representations with trainable parameters W^e. Starting from x, the feature extractor generates the original feature representation $h_{all} \in \mathbb{R}^{F_1 \times 1}$, which can be formulated as,

$$h_{all} = \mathcal{E}(x; W^e) \tag{2}$$

Here, the feature extractor $\mathcal{E}$ is not limited to a specific model, and most existing single-domain models can be integrated into our LIMFA as a feature extractor by removing their classification layers. Accordingly, input x is determined by the specific feature extractor utilized. The feature extractor generates the feature representation $h_{all} \in F_1 \times 1$, which is common in all base models. For example, in BiGCN w/LIMFA, the bi-directional graph convolutional layers [1] are employed as the feature extractor, and the input x is the sentence embedding of the news claim and its comments, and diffusion structure.

Unlike previous studies that imposed alignment constraints on the original features of different domains [13, 34], we enforced the alignment on the adaptively selected features to find the subspace that is suitable for alignment and conducive to disentanglement, respectively. Inspired by the channel attention [7], we design an elementwise attention gate g to separate the original feature representation into two complementary parts with dimension signal values ranging from 0 to 1. That is, the value of the elementwise attention gate g is decided by the original feature representations h_{all} with a perception layer followed by a sigmoid function. Afterward, performing elementwise product of h_{all} with g and $1 - g$, respectively,

$$g = sigmoid(h_{all}W^g + b^g) \tag{3}$$

$$h_{di} = g \circ h_{all} \tag{4}$$

$$h_{ds} = (1 - g) \circ h_{all} \tag{5}$$

where $W^g \in \mathbb{R}^{F_1 \times F_1}$ and $b^g \in \mathbb{R}^{F_1}$ are the gate parameters. By gating the original features, we separate two feature sub-spaces, $h_{di} \in \mathbb{R}^{F_1 \times 1}$ and $h_{ds} \in \mathbb{R}^{F_1 \times 1}$, and the complementarity between them is conducive for disentanglement. Further supervision is needed to ensure two feature subspaces that one being invariant across domains and the other providing discriminative information other than DIF, which is detailed in Sects. 3.4 and 3.5. With the help of loss functions, the gate g enables the features h_{all} to be clearly disentangled into DSF h_{ds} and DIF h_{di}.

Restorer To prevent the model from achieving the goal of feature alignment by discarding as many features as possible, we design the restitution loss to monitor the information loss during feature extraction and disentanglement. In detail, the restorer $\mathcal{D}$ first restitutes the sample from the separated features and the restitution loss then measures the information loss between the original sample and the restituted sample. The restorer $\mathcal{D}$ is set to the reverse order of the feature extractor $\mathcal{E}$, which conforms to the encoder–decoder architecture. By feeding the decoder with separated features h_{di} and h_{ds}, we obtain the restituted sample $x_{res} \in \mathbb{R}^{F_0 \times 1}$, that is,

$$x_{res} = \mathcal{D}(h_{di} + h_{ds}; W^{d}) \tag{6}$$

The more similar the restituted data and corresponding original data are, the less the information loss will be. Therefore, the restitution loss is performed on x_{res} and x as,

$$\mathcal{L}_{res} = \frac{1}{N}\sum_{i=1}^{N} \|x^{i}_{res} - x^{i}\|_1 \tag{7}$$

where N denotes the number of samples, $\|\cdot\|_1$ is the averaged L1-distance over the number of posts and feature dimensions.

Classifiers In the final prediction, we employ two classifiers, in which the main classifier relied on DIF is transferred to the target domains while the auxiliary classifier relied on DSF features helps the feature disentanglement. Both of them deploy a fully connected layer with softmax to predict whether the sample is fake news or not. The main classifier $\mathcal{C}_{di}$ takes the DIF h_{di} as input, which can be formulated as,

$$\hat{y}_{di} = \mathcal{C}_{di}(h_{di}; W^{di}, b^{di}) \tag{8}$$

where $W^{di} \in \mathbb{R}^{F_1 \times 1}$, $b^{di} \in \mathbb{R}$ are the parameters. The cross-entropy loss is employed to train it,

$$\mathcal{L}_{ce_{di}} = -\frac{1}{N}\sum_{i=1}^{N} y^{i} \log y^{i}_{di} + (1 - y^{i}) \log(1 - y^{i}_{di}) \tag{9}$$

where y^{i} and $\hat{y}^{i}_{da}$ denote the ground truth label and the predicted label of the i-th sample, respectively, and N denotes the total number of samples.

News pieces of different domains have their own unique or specific features that offer information for making more precise predictions [25]. In other words, after adding the DSF to the DIF, the overall features should be more discriminative, resulting in a less ambiguous predicted category with a higher likelihood. Therefore, the DSF can be identified by measuring the information gain they bring to each news piece. Based on this, the auxiliary classifier $\mathcal{C}_{all}$ is designed to learn the DSF for a fixed DIF. In this way, we can estimate the information gain of the DSF easily by the likelihood difference between the auxiliary classifier and the main classifier. Formally, the auxiliary classifier fed with the fixed DIF h_{di} and DSF h_{ds} is,

$$\hat{y}_{all} = \mathcal{C}_{all}(h_{di} + h_{ds}; W^{all}, b^{all}) \tag{10}$$

where $W^{all} \in \mathbb{R}^{F_1 \times 1}$ and $b^{all} \in \mathbb{R}$ are the parameters. The fixed DIF h_{di} means it is isolated from the back-propagation of the auxiliary classifier. We employ the cross-entropy loss to achieve the goal that the DSF is complementary to the DIF and useful for classification,

$$\mathcal{L}_{ce_{all}} = -\frac{1}{N}\sum_{i=1}^{N} y^{i} \log y^{i}_{all} + (1 - y^{i}) \log(1 - y^{i}_{all}) \tag{11}$$

where y^{i} and $\hat{y}^{i}_{all}$ denote the ground truth label and the predicted label of the i-th sample, respectively, and N denotes the total number of samples. Further learning on DSF by information gain is detailed in Sect. 3.5.

3.4 Center-aware feature alignment

Despite the DIF getting rid of DSF, there is no guarantee that the DIF is shared by all source domains. We now describe the center-aware feature alignment component of our LIMFA. When the target domain can be accessed, previous studies [12, 34] construct data pairs in each mini-batch that sample data from both source and target domains, respectively. They take the distance between data pairs as the alignment constraint to minimize the distribution divergence of the source and target domain. For our problem, a straightforward solution is to construct data pairs for the M source domains and minimize the $M(M-1)$ pairs of the distance between domains simultaneously. However, such a solution is computationally intensive and requires domain labels for all training samples.

Instead, we learn the category center and minimize the distance between samples and the center to minimize the distribution divergence of source domains. Intuitively, aligning the multiple domains simultaneously is equal to aligning them to their center. Therefore, we take the center loss [31] as the reference and propose the likelihood-based center loss. The center loss averages the feature representation of the corresponding categories in each mini-batch as centers and minimizes the distance between intra-category samples and their centers to reduce the feature distribution divergence. Similarly, we perform the mini-batch training and shuffle the samples to ensure a more uniform domain distribution for each category per batch. Considering samples of the same domain but different categories may perturb the separation between category centers, we further take classification results into computation to push centers away from the classification boundary, thus keeping the feature distribution of different categories separated in the embedding space. Hence, the modified centers are formulated as,

$$c_y = \frac{\sum_{i=1}^{N} \mathbb{1}_y^i * \hat{y}_{di}^i * h_{di}^i}{\sum_{i=1}^{N} \mathbb{1}_y^i * \hat{y}_{di}^i}$$
$$c_{1-y} = \frac{\sum_{i=1}^{N} \mathbb{1}_y^i * (1-\hat{y}_{di}^i) * h_{di}^i}{\sum_{i=1}^{N} \mathbb{1}_y^i * (1-\hat{y}_{di}^i)} \quad (12)$$

where N is the number of samples, $c_y \in \mathbb{R}^{F_1 \times 1}$ and $c_{1-y} \in \mathbb{R}^{F_1 \times 1}$ are the centers of two categories. $\mathbb{1}_y$ and $\mathbb{1}_{1-y}$ are the indicators of mismatched samples for two categories, $\mathbb{1} \in (0,1)_1^N$ takes value 1 when the prediction category matches the label, and 0 otherwise. The category centers are biased toward features with higher likelihood $\hat{y}_{di}$ to enlarge the inter-category distance. Then, our likelihood-based center loss can be formulated as follows,

$$d(h_{di}, c) = \|h_{di} - c\|_2 \quad (13)$$

$$\mathcal{L}_{cc} = \frac{1}{N}\sum_{i=1}^{N}(\mathbb{1}_y^i d_y^i + \mathbb{I}_{1-y}^i d_{1-y}^i) \quad (14)$$

where $d(\cdot,\cdot)$ employs L2-distance. By minimizing the $\mathcal{L}_{cc}$, the correctly classified samples are pulled to category centers, which enforces the potential domain centers to be aligned to the corresponding category centers, minimizing the domain differences.

3.5 Likelihood gain-based feature disentanglement

We now describe the likelihood gain-based feature disentanglement component of our LIMFA. As aforementioned, further supervision is needed to identify the DSF and disentangle them from the aligned DIF. We design likelihood gain loss $\mathcal{L}_{lg}$ to encourage the DSF to preserve more information other than DIF. Since the DIF and the DSF are complementary to each other, the more informative the DSF is, the less informative the DIF is, such that the DSF is disentangled from the DIF competitively. The difference between the prediction likelihood of two classifiers is utilized to approximate the information gain of DSF, which is defined as follows:

$$lg = y(\hat{y}_{di} - \hat{y}_{all}) + (1-y)(\hat{y}_{all} - \hat{y}_{di}) \quad (15)$$

$$\mathcal{L}_{lg} = \frac{1}{N}\sum_{1=1}^{N}\max(lg^i, 0) \quad (16)$$

where the $\hat{y}_{di}$ and $\hat{y}_{all}$ are the prediction labels of two classifiers. The $lg \in [-1,1]$ is the likelihood gain of the DSF, where $lg > 0$ indicates the DSF has no more information other than DIF and thus needs to be penalized. To ensure stable training, this penalty term is truncated when $\mathcal{L}_{lg} < 0$. Here, the $\hat{y}_{di}$ is isolated from backpropagation to encourage the DSF to preserve more information rather than extort DIF.

3.6 Integrated loss function

The components in LIMFA update their parameters simultaneously using the integrated loss function $\mathcal{L}_{all}$. Formally, our ultimate goal is to optimize the following,

$$\mathcal{L}_{all} = \mathcal{L}_{res} + \mathcal{L}_{ce_{all}} + \mathcal{L}_{lg} + \mathcal{L}_{ce_{di}} + \alpha\mathcal{L}_{cc} \quad (17)$$

where the parameter α defines the importance assigned to likelihood-based center loss item during training phase. To prevent the features from being aligned when they are uncertain, the α is adjusted by the prediction likelihood of the main classifier. Specifically,

$$\alpha = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbb{1}_y^i \hat{y}_d i^i + \mathbb{1}_{1-y}^i (1-\hat{y}_{di}^i)\right) \quad (18)$$

$$\alpha = \begin{cases} 0, \alpha \le 0.5 \\ \alpha, otherwise \end{cases} \quad (19)$$

The adaptive parameter α is truncated when the average prediction likelihood is under 0.5. Otherwise, it equals the prediction likelihood.

4 Experiments

To achieve a comprehensive evaluation, we performed six experiments to simulate fake news detection in real applications. For the effectiveness evaluation, we compare the performance of single-domain methods with/without our LIMFA framework in Sect. 4.2, and the performance of cross-domain models in Sect. 4.3. For robustness evaluation, we simulate the multiple target domains in Sect. 4.4, the low-correlated target domain in Sect. 4.5 and multiple source domains in Sect. 4.6. In Sect. 4.7, we evaluate the contributions of each component to our LIMFA framework. To verify the issues raised in the Introduction, we visualize the feature distribution of source domains and source vs. target domain to conveniently observe the difference between LIMFA and existing baselines.

4.1 Setup

4.1.1 Datasets

Experiments are constructed on two widely used datasets, Pheme (mainly in English) and Weibo (in Chinese), which have natural domain labels according to their sampling mechanisms. The Pheme [40] is collected from Twitter relating to 5 newsworthy events, which is labeled by event. The Weibo21 [19] dataset collects new pieces of nine topics on Sina Weibo, including the Science, Military, Education, Disaster, Politics, Health, Finance, Entertainment, and Society domains. We denote the full dataset as Ch-9. For each news piece, both two datasets record the original post and all its repost/reply posts. The Pheme dataset further collected the reply relationship between posts. The detailed statistics are listed in Table 2.

We visualize the domains under the domain classification task using t-SNE to report the relationships between domains. The domain boundaries of events in the Pheme dataset are clear. The low correlations between the Ferguson event and other events imply a challenging cross-domain detection. The same goes for the Germanwings Crash event. In contrast, the domain boundaries of nine domains in the Weibo dataset are fuzzy, especially the Society domain, which involves almost every other domain. Such correlations inspire us to remove some domains from the Weibo dataset to test the robustness of baselines in learning DIF from less correlated domains. Therefore, we sample two datasets from the Weibo dataset as Ch-3 and Ch-6. Ch-3 contains the more correlated domains, including Finance, Entertainment, and Society domains. Ch-6 contains the rest 6 domains.

4.1.2 Implementation details

All the baselines are implemented with Pytorch. We use pre-trained Bert models [32] followed by a mean-pool to generate the sentence embedding vectors as input x. For single-domain models, the extra inputs (diffusion structure) are processed on request, and the models are reassembled in encoder–decoder architecture to deploy in our LIMFA. Besides, we adopted a unified set of hyper-parameters throughout the Weibo and Pheme datasets, where $batch-size = 64$, and the max number of responsive posts $m = 50$. The learning rate is initialized as $5e-4$ and gradually decreases during the process of training. We do not otherwise perform any dataset-specific tuning other than early stopping on dev sets.

Table 2 Statistics of datasets

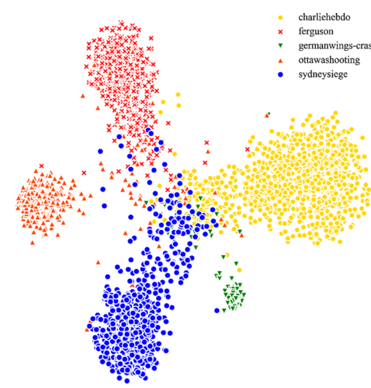
Dataset	Domains	ALL	Fake	Real	Visualizations of domains
Pheme	Charlie Hebdo (C)	2079	458	1621	
	Ferguson (F)	1143	284	859	
	Germanwings Crash (G)	469	238	231	
	Ottawa Shooting (O)	890	470	420	
	Sydney Siege (S)	1221	522	699	
Weibo	Science (Sci.)	236	93	149	
	Military (Mili.)	343	222	121	
	Education (Edu.)	491	248	243	
	Disasters (Disa.)	776	591	185	
	Politics (Poli.)	852	546	306	
	Health (Heal.)	1000	515	485	
	Finance (Fina.)	1321	362	959	
	Entertainment (Ent.)	1440	440	1000	
	Society (Soc.)	2669	1471	1198	

Table 3 The architecture of the baselines

Models	Layers and Parameters					
Text-CNN	-	Conv1 (1, 768)×5	Conv2 (2,768)×5	Conv3 (3,768)×5	Conv4 (4, 768)×5	FC (4,200)(200,2)
BIGCN	Text-EMB (768,200)	BU-GCN-1 (400,200)	BU-GCN-2 (400,200)	TD-GCN-1 (400,200)	TD-GCN-2 (400,200)	FC (800,2)
ClaHi-GAT	Text-EMB (768,200)	GAT-1 (200,200)+2 m	GAT-2 (400,200)+2 m	NLI (800,200)	Attention (200,1)	FC (400,2)
Trans-E (LIMFA)	Text-EMB (768,200)	Self-Attention (400,200)×3	FFN (200,200)	Post-Attention (400,1)	Gate (200,200)	FC1, 2 (200,2)
EANN	Conv1 (1, 768)	Conv2 (2,768)	Conv3 (3,768)	Conv4 (4, 768)	Discriminator (200, 200)(200,4)	FC (4, 200)(200, 2)
BDANN	Text-EMB (768,200)	Self-Attention (400,200)× 3	FFN (200,200)	-	Discriminator (200,200)(200,4)	FC (4,200)(200,2)
MDFEND	Conv[1-5] ×Expert $[([1, 2, 3, 5, 10], 768) \times 64] \times 3$		MaskAttention (768, 1)	Domain-Emb (*num_Dom*,200)	Gate (1536,200)(200,3)	MLP (320,200)(200,2)
M3FEND	Conv[1-5] × num_Dom $[([1, 2, 3, 5, 10], 768) \times 64] \times num\_Dom$		Gate (1536,200)(200,200)	Domain-Emb (*num_Dom*,768)	Domain-Memory (768, 768) × 2	MLP (320,200)(200,2)
EDDFN	-	Domain-Inv (768,200)	Domain-Spe ×*num_Dom* (768, 200) ×*num_Dom*		Decoder (400, 200)(200,768)	MLP (400,2)

Following previous work, the leave-one-domain-out cross-validation (LOO-CV) [17] and k-fold cross-validation (fivefold CV) [19] are employed to evaluate the generalization ability of LIMFA for source domains and target domains, respectively. The performance is mainly characterized by the accuracy (ACC.) and macro F1 score. The architecture and detailed parameter size of the baselines are listed in Table 3.

4.1.3 Comparison models

The first group is the models designed to improve performance for a specific domain, marked as **G1**. We choose four representative single-domain models as base models, including Text-CNN [9], BIGCN [1], ClaHi-GAT [12] and the Trans-E(A transformer encoder–decoder) [28]. The Text-CNN and Trans-E purely focus on text features, while the other two baselines further improve performance by diffusion structure and implicit stance, respectively. We give a brief overview of all baselines as follows.

- Text-CNN: A classical sentence classification model, that is composed of one layer of convolution (4 convolving filters in different sizes [1, 2, 3, 4]).
- BIGCN: A novel bi-directional graph model to explore characteristics in both bottom-up propagation and top-down dispersion.
- ClaHi-GAT: A graph attention network(GAT)-based model that enhances the representation learning for responsive posts considering the entire social contexts and attends over the posts that can semantically infer the target claim.
- Trans-E: A transformer encoder–decoder with root enhancement and a classifier.

The second group is the models designed to improve generalization for multiple source domains, marked as **G2**, including the MDFEND [19], EDDFN[25] and M3FEND [39].

- EDDFN: EDDFN models DIF and DSF simultaneously with MLPs.
- MDFEND: A mixture of experts is proposed to learn the multiple representations adaptively. Meanwhile, a domain gate generated from domain embedding aggregates the multiple representations.
- M3FEND: The latest multi-domain fake news detection model leverages the domain-memory bank and domain adapter to preserve features for each domain. For comparison, we only use the content view.

The third group is the models designed to improve the generalization for target domains, marked as **G3**. In this category, we only find the EANN [30] and BDANN [37], which adopt the domain discriminator to preserve the DIF at the output level. Different from our LIMFA, both of them need domain labels for training a domain discriminator. The details are as follows:

- EANN: EANN employs a domain (event) discriminator that is implemented with a gradient reversal layer to remove DSF and keep shared features among news domains.
- BDANN: BDANN replaces the feature extractor TextCNN in EANN with transformer.

In this paper, we remove the visual information module of baselines to suit the problem settings.

4.2 Experiments on the G1 group

In this section, we evaluate our LIMFA over G1 to illustrate its effectiveness and robustness. Since BIGCN and ClaiHi-GAT require the diffusion structure, and Weibo21 does not satisfy this requirement, we only evaluate them on Pheme. The performance of these base models within and without LIMFA framework is shown in Table 4. * ($p \leq 0.05$) indicates paired t test of LIMFA vs. the best baseline. Bold results indicate the best. We have the following observations.

(1) In Table 4, all baselines achieve better performance after being migrated in our LIMFA, improving by 3.87 in accuracy and 4.69 in F1 on average, which illustrates that our LIMFA strengthens the generalization ability of models on target domains.

(2) For five events, the improvement in the C, O and S events is more significant than that in F, and G. We attribute such a difference between the five events to the correlations between domains. Compared to the F and G events, the C, O and S events are more correlated, implying there may be more DIF that are shared among them and thus get more improvements from others.

4.3 Experiments on the G2 and G3 Groups

In this section, we conduct experiments to compare the G2 and G3 groups. The results of Pheme and Weibo CH-9 are reported in Tables 5 and 6, respectively (The ACC of Table 6 are listed in Appendix.). *($p \leq 0.05$) indicates the paired t test of LIMFA vs. BDANN. Bold and underlined results indicate the best and second best, respectively. We have the following observations,

(1) Comparing Tables 4 and 5, EANN outperforms TextCNN by introducing a domain discriminator. Similar results can be found between BDANN and Trans-E. This validates the effectiveness of aligning feature distribution to enhance model generalization.

(2) The EANN, BDANN and LIMFA outperform MDFEND and EDDFN on most tasks. The MDFEND and EDDFN aim to improve the performance of all domains via modeling the domain knowledge of each source domain. Therefore, it exhibits an impaired performance when the news pieces come from a target domain that has no related domain knowledge. Instead, the EANN, BDANN and our LIMFA strive for the DIF that are shared by multiple domains and thus are well transferred to the target domain. Such phenomena demonstrate the necessity of fake news detection targeting potentially unseen domains.

Table 4 Result of LOO-CV on the Pheme dataset(G1)

	Method	Charlie Hebdo		Ferguson		Germanwings		Ottawa shooting		Sydney Siege	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G1	BIGCN	0.7572	0.7038	0.7166	0.6347	0.6088	0.6009	0.6313	0.6173	0.6893	0.6487
(G3)	w/ LIMFA	**0.7644**	**0.7273**	**0.7166**	**0.6381**	**0.6294**	**0.6157**	**0.7634**	**0.7659**	**0.7380**	**0.7170**
	ClaHi-GAT	0.7974	0.7410	0.6964	0.5830	0.5383	0.5074	0.6342	0.6141	0.7106	0.6845
	w/ LIMFA	**0.8147**	**0.7539**	**0.7227**	**0.6190**	**0.6176**	**0.6263**	**0.7137**	**0.7398**	**0.7167**	**0.6994**
	TextCNN	0.8194	0.7444	0.7064	**0.6489**	0.5441	0.5318	0.6700	0.6597	0.6927	0.6491
	w/ LIMFA	**0.8276**	**0.7492**	**0.7429**	0.6470	**0.5765**	**0.5731**	**0.7242**	**0.7239**	**0.7175**	**0.7022**
	Trans-E	0.7631	0.7377	0.7288	0.6320	0.6206	0.6203	0.6879	0.6804	0.7044	0.6687
	w/ LIMFA	**0.8240***	**0.7623***	**0.7479***	**0.6433***	**0.6265***	**0.6210***	**0.7287***	**0.7270***	**0.7397***	**0.7218***

Table 5 Results of LOO-CV on the Pheme dataset (G2 and G3)

	Method	Charlie Hebdo		Ferguson		Germanwings		Ottawa shooting		Sydney Siege	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G2	EDDFN	0.8222	0.6785	0.7119	0.5630	0.5735	0.5637	0.7018	0.6932	0.6523	0.6196
	MDFEND	0.7838	0.7011	0.7088	0.6164	0.5441	0.5435	0.6452	0.6323	0.7094	0.6732
G3	EANN	0.7970	0.7456	0.7150	0.6261	0.5912	0.5904	0.7008	0.6945	0.7313	0.7249
	BDANN	0.8081	0.7468	0.7242	0.6385	0.5883	0.5877	0.7247	0.7173	**0.7509**	**0.7418**
	Ours	**0.8240***	**0.7623***	**0.7479***	**0.6433***	**0.6265***	**0.6210***	**0.7287**	**0.7270**	0.7397	0.7218

Table 6 Results of LOO-CV on the CH-9 dataset (G2 and G3)

	Method	Fina.	Edu.	Mil.	Sci.	Soc.	Ent.	Heal.	Disa.	Poli.
G2	EDDFN	0.8452	0.8562	**0.8617**	0.8405	0.8158	0.8531	0.8563	0.8434	0.8591
	MDFEND	0.8442	0.8411	0.8191	0.8387	0.8326	0.8600	0.8294	0.8091	0.8377
G3	EANN	0.8620	0.8585	0.8232	0.8400	0.8553	0.8499	0.8541	0.8364	0.8536
	BDANN	0.8498	0.8563	0.8384	0.8354	0.8623	0.8607	0.8586	**0.8587**	0.8614
	Ours	**0.8732***	**0.8676***	0.8423	**0.8424***	**0.8695**	**0.8683**	**0.8741***	0.8439	**0.8710***

(3) Comparing Tables 5 and 6, the detection results in the Weibo dataset are better than that in the PHEME dataset. The reason is twofold, sufficient data and higher correlation between domains, which may provide more potential common features to the target domain.

(4) With the results of the t test, we find that LIMFA outperforms the best baseline significantly in most tasks, which demonstrates that LIMFA is an effective solution to improve generalization for unseen target domains. Two improvements contributed to such achievements, the distribution alignment at the feature level and enhanced feature disentanglement, enhancing the robustness of DIF across all domains. Moreover, LIMFA does not need to annotate the domain for news pieces, and that is the greatest strength of our solution.

4.4 Generalizing to multiple target domains

Since real news streams may comprise more than one domain that is unseen in the training dataset, in this section, we conduct experiments to simulate such a scenario. Models are trained on the CH3 dataset and tested on the CH6 dataset to verify their robustness when detecting fake news from multiple unseen domains. The results are listed in Table 7. We have the following observations:

(1) Our LIMFA framework outperforms other baselines on overall detection performance, which demonstrates that LIMFA is an effective solution to improve the generalization of models to any unseen domain.

(2) Compared to Table 6, the performance of all baselines drops slightly in most domains due to the reduction in the number of source domains. Surprisingly, the performance degradation is not as severe as we thought. The reason could be that these three source domains are highly correlated with target domains, especially the Society domain whose feature distribution overlaps most of the others (see Sect. 4.7).

(3) Compared to Table 6, the improvement of our LIMFA over the others is not that obvious as the number of source domain decreases. In other words, our framework widens the gap with the baseline when the number of source domains increases, demonstrating the effectiveness of likelihood-based center loss in aligning multiple domains.

4.5 Generalizing to less-correlated target domains

In this section, experiments are conducted to evaluate the robustness of baselines when the correlations between the source and the target domain are low. In such scenarios, learning the DIF that generalize well to news samples of target domains becomes significantly challenging. We construct LOO CV on the CH-6 dataset, and the summarized results are presented in Table 8. We report our observations:

Table 7 Results of models that are trained on the CH3 dataset and tested on the CH6 dataset(multiple unseen target domains)

	Method	Edu.		Mil.		Sci.		Heal.		Disa.		Poli.		Overall	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G2	EDDFN	0.8181	0.8130	0.8376	0.8210	0.8549	0.8437	0.8561	0.8558	0.8416	0.8062	0.8325	0.8251	0.8401	0.8275
	MDFEND	0.8051	0.8035	0.8164	0.8037	0.8384	0.8182	0.8063	0.8057	0.8187	0.7828	0.8170	0.8103	0.8170	0.8040
G3	EANN	0.8502	0.8497	0.8492	0.8367	0.8625	0.8303	0.8561	0.8560	0.8428	0.8084	0.8375	0.8285	0.8497	0.8349
	BDANN	0.8421	0.8398	0.8410	0.8287	0.8687	0.8361	0.8684	0.8684	0.8543	0.8221	0.8539	0.8459	0.8548	0.8401
	Ours	**0.8551**	**0.8535**	**0.8523**	**0.8382**	**0.8694**	**0.8398**	**0.8753**	**0.8753**	**0.8667**	**0.8317**	**0.8655**	**0.8572**	**0.8640**	**0.8493**

(1) When removing Society, Entertainment and Finance domains from the training dataset, the performance of EANN and BDANN notably declines across most domains. Such spotty performance across domains suggests that the learned DIF lack sufficient generalization, validating our earlier analysis in Sect. 1 that incomplete domain labels lead to incomplete feature alignment and feature entanglement.

(2) Models exhibit inconsistent performance across domains. In the Science, Disaster, and Politics domains, the multi-domain baselines demonstrate superior performance, whereas in the remaining domains, the cross-domain baselines appear to be more effective. This variability indicates the complex correlations between domains.

(3) Among all baselines, our LIMFA performs best, which indicates that based likelihood-based center loss is more effective in learning multi-domain DIF, even in scenarios with low inter-domain correlations.

4.6 Generalizing to multiple source domains

In this section, we evaluate the influences of DIF on the discrimination of models. For this purpose, we pooled all news domains and split the dataset into train/evaluation/test sets in a ratio of 8:1:1. Table 9 reports the results of the unseen-domain and multi-domain models. Unlike the previous sections, we report the results of the auxiliary classifier in our model, denoted as Ours-ALL.

From this task, we found that the performance of the DIF-only model is slightly worse than that of the model using DIF and DSF together, but this gap can be offset by the robust backbone (e.g., Transformer Encoder). For the Pheme dataset, the EDDFN, MDFEND and M3FEND outperform the EANN but are slightly worse than BDANN. As for the Weibo dataset, the EDDFN achieves sub-optimal performance and outperforms the BDANN, which indicates that the domain-specific features bring additional discrimination to fake news detection, especially when the training data contain many news domains. Our LIMFA-ALL outperforms all baselines, including LIMFA, which demonstrates that preserving domain-specific features is necessary for source domains, but fine-grained design is needed when transfer to target domains.

4.7 Ablation study

To demonstrate the effectiveness of the proposed likelihood gain-based feature disentanglement and center-aware feature alignment components, we carry out ablation experiments in this section. The contributions of

Table 8 Results of LOO-CV on the CH-6 dataset (the less-correlated domains)

	Method	Edu.		Mil.		Sci.		Heal.		Disa.		Poli.	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G2	EDDFN	0.8269	0.8251	0.8517	0.8483	0.8259	0.8203	0.7926	0.7909	0.8445	0.8166	0.8538	0. 8434
	MDFEND	0.7586	0.7585	0.7697	0.7623	0.8304	0.8210	0.7540	0.7507	0.8423	0.7979	0.7542	0.7517
G3	EANN	0.8287	0.8275	0.8218	0.7772	0.7856	0.7746	0.7953	0.7948	**0.8670**	0.8103	0.8357	0.8243
	BDANN	0.6468	0.6465	0.8265	0.8171	0.7991	0.7910	0.7254	0.7010	0.8601	0.8175	0.6132	0.3817
	Ours	**0.8330**	**0.8324**	**0.8565**	**0.8417**	**0.8371**	**0.8298**	**0.8387**	**0.8384**	0.8553	**0.8192**	**0.8558**	**0.8454**

components in our LIMFA to better transferability are listed in Tables 10 and 11.

We observed that removing the feature disentanglement component (w/o FD) of LIMFA damages the performance severely, followed by removing the feature alignment components (w/o FA), which indicates that feature disentanglement and feature alignment are both needed for learning the DIF. And the sharp performance degradation of w/o FD indicates that the direct alignment on overall feature representations may impair the discrimination of the model when the source domain contains many different news domains. By, respectively, removing $\mathcal{L}_{lg}$, $\mathcal{L}_{res}$, $\mathcal{L}_{cc}$ and the likelihood of the center loss from the best-performing LIMFA framework, we see the performance decrease, indicating these loss functions are all necessary and play complementary roles in detection. The performance drops when removing the likelihood gain loss function, indicating some DSF are entanglement with DIF and only identifying the DIF is insufficient. The same goes for the likelihood-based center loss, which demonstrates that expanding the distance between different categories is necessary to counteract the correlation of the intra-domain news pieces. Additionally, removing the restitution loss function leads to a slight performance drop, which confirms the necessity of restitution loss in preventing information loss during feature disentanglement. Due to the complex correlation between source and target domain data, the loss functions perform not always present consistent results across all datasets. But every component has its effects on the optimal and robust LIMFA.

4.8 Visualizations

To provide an intuitive understanding of what the proposed LIMFA is doing, we visualize feature distributions of multiple source domains, and feature distributions of source and target domain, using t-SNE [27] projection. All models perform LOO-CV on the CH-9 dataset and the Science domain is isolated for testing.

For the multiple source domains, we select the society domain as the reference and visualize other domains. In Fig. 3, we can see that (a) Trans-E: without the treatment to domain discrepancy. Samples within a domain are clustered obviously and samples of different domains are scattered over separated spaces. For instance, some samples of the Finance domain are clustered far away from the main cluster of the Society domain, and they do not share spaces with other domains. (b)EANN and (c) BDANN: leverages domain discriminator to preserve DIF in the representation space. Samples are pulled closer yet not aligned across all source domains. For example, the separated cluster of Finance domains in Trans-E is alleviated,

Table 9 Results of fivefold CV

	Method	Weibo				Pheme			
		ACC	Prec	Rec	F1	ACC	Prec	Rec	F1
G2	EDDFN	0.9080	0.9084	0.9075	0.9078	0.7532	0.7867	0.7312	0.7106
	MDFEND	0.8785	0.8686	0.8524	0.8584	0.7469	0.7482	0.6946	0.6932
	M3FEND	0.8924	0.8809	0.8777	0.8775	0.7508	0.7443	0.7073	0.7063
G3	EANN	0.8912	0.8826	0.8697	0.8750	0.7420	0.7388	0.7340	0.7142
	BDANN	0.9012	0.8901	0.8859	0.8872	0.7853	0.7691	0.7670	0.7558
	Ours	0.9032	0.8875	0.8753	0.8796	0.7810	0.7615	0.7397	0.7427
	Ours-ALL	0.9100	0.9070	0.8921	0.8974	0.7914	0.7737	0.7519	0.7549

Table 10 The results of ablation study (Pheme dataset). Each row indicates the relative improvements (%) compared with our LIMFA framework

Method	Charlie Hebdo		Ferguson		Germanwings		Ottawa shooting		Sydney Siege	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
w/o FD	−1.40	−1.36	−5.74	−1.05	−1.76	−1.66	−2.49	−2.52	1.40	1.54
w/o FA(L_{cc})	−1.33	−2.14	−0.14	−3.74	−0.41	−0.07	−4.07	−4.06	−4.24	−3.68
w/o FA+FD	−6.09	−2.46	−1.91	−1.13	−0.59	−0.07	−4.08	−4.66	−3.53	−5.31
w/o L_{lg}	0.25	0.00	−5.33	−0.63	−0.88	−1.28	−0.61	−0.56	1.06	1.54
w/o L_{res}	−2.73	−2.12	−5.88	−0.47	−0.11	−0.16	0.005	0.04	−0.72	−0.13
w/ center	0.47	0.55	−5.33	−1.25	−0.29	−0.37	−0.71	−0.82	1.01	1.59

Table 11 The results of ablation study (Ch-3 to CH-6 dataset). Each row indicates the relative improvements (%) compared with our LIMFA framework

Method	Edu.		Mil.		Sci.		Heal.		Disa.		Poli.	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
w/o FD	−16.32	−16.44	−6.36	−8.58	−6.52	−8.51	−13.03	−13.11	−14.65	−18.03	−12.55	−15.42
w/o FA(L_{cc})	−0.68	−0.80	−2.10	−1.91	0.40	0.51	−0.39	−0.40	−0.73	−0.57	−1.75	−1.79
w/o FA+FD	−2.49	−2.75	−3.37	−3.08	−2.73	−3.43	−1.29	−1.30	−2.10	−2.25	−2.52	−2.41
w/o L_{lg}	−2.39	−2.53	−1.95	−1.78	−1.16	−1.30	−0.29	−0.28	−1.55	−1.67	−0.91	−0.87
w/o L_{res}	−0.36	−0.26	−2.89	−3.58	−1.61	−1.19	−0.65	−0.69	0.23	−0.34	−1.04	−1.51
w/ center	0.59	0.70	−0.84	−1.45	−0.94	−0.63	−1.02	−1.07	−0.80	−1.56	−1.36	−1.81

it overlaps with the Health domain and the Politics domains. However, we observe a clear distribution discrepancy between the Entertainment domain and the Finance domain though they both overlap with the Society domain. Such phenomena prove our analysis of the incomplete feature alignment under domain discriminator. (d) LIMFA: leverages contrastive learning to preserve DIF in the representation space. The clustering phenomenon of domains has been alleviated, and samples of different domains are nearly uniformly distributed over the representation space. Moreover, different categories of features are separated clearly in our LIMFA, indicating that the likelihood-based center loss reduces the intra-domain similarity while enlarging the invariance within the category.

From Fig. 4, we observe that (a) Trans-E: shifts are observed between the source and target domain distributions; (b) EANN and (c) BDANN: the samples within the

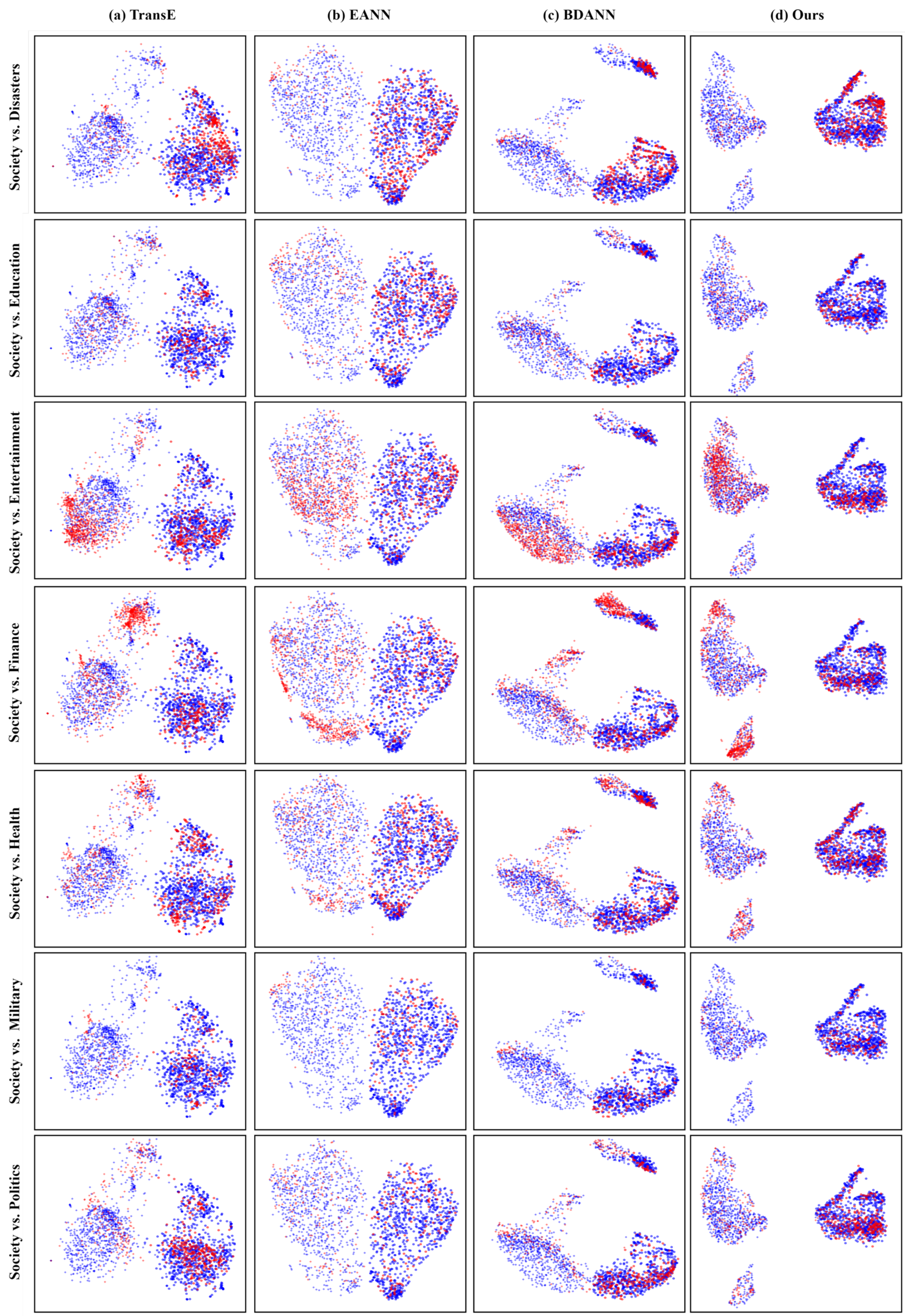


Fig. 3 (Best viewed in color.) The t-SNE visualization of source domain space generated by Trans-E (**a**), EANN (**b**), BDANN (**c**) and LIMFA (**d**). The domains are distinguished in colors, and the news and fake news are marked with circle and triangle, respectively (Color figure online))

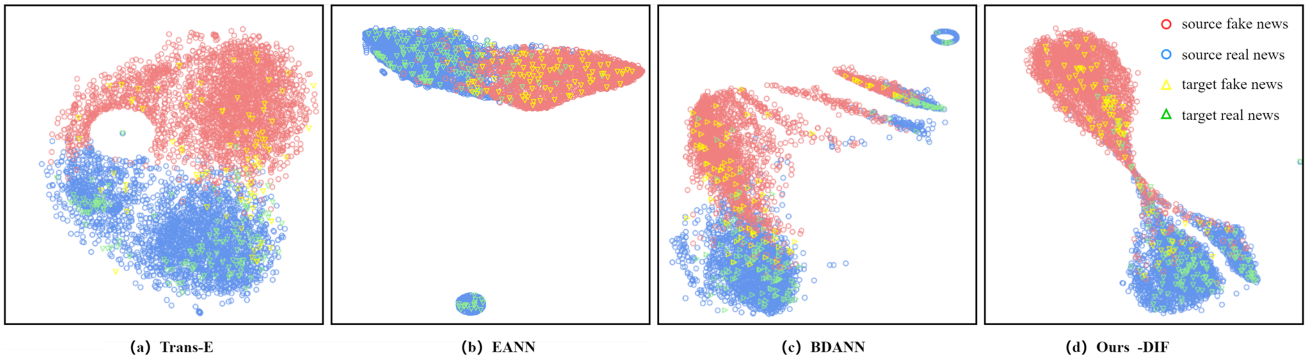


Fig. 4 (Best viewed in color.) The t-SNE visualization of the source and target domain space generated by Trans-E (**a**), EANN (**b**), BDANN (**c**) and LIMFA (**d**). The fake news and real news are marked with circle and triangle, respectively (Color figure online))

source domain have multiple sub-clusters and some do not overlap with the target domain. Such phenomena can be attributed to incomplete feature alignment that some DSF are preserved and cannot be transferred to the target domain; and (d) LIMFA: the features of source and target domains are concentrated to the category centers with a good category-wise mix compared to the baselines. These observations provide strong support for the effectiveness of center-aware feature alignment and feature disentanglement and likelihood gain-based feature alignment.

5 Conclusion and discussions

This paper focuses on improving the generalization to the target domain while maintaining the discrimination on the source domain. To achieve this objective, we analyzed the necessity of preserving the domain-specific features (DSF) and identified two challenges in learning domain-invariant features (DIF), complex inter-domain correlations and high inter-domain correlations. Based on these findings, we proposed a label-irrelevant multi-domain alignment framework. The framework consists of two key components, the center-aware feature alignment component, which enhances generalization across domains, and the likelihood gain-based feature disentanglement component, which promotes the better disentanglement of domain invariant/specific features. Notably, our LIMFA does not use domain labels, making it easily implementable. Through experiments conducted on real-world datasets, we found that LIMFA serves as a simple yet effective tool for improving the generalizability of base models. Specifically, our experiments demonstrated that LIMFA outperforms existing methods in two scenarios: when there are numerous source domains available for training and when the target domain significantly differs from the source domains.

However, there are areas for further exploration. Firstly, although the DSF is preserved in our approach, they are not utilized for detecting fake news in the target domains, potentially resulting in the loss of useful information. As aforementioned, the correlations between news samples are complex, and certain DSF may be useful for some news samples but not for others. Therefore, we suggest conducting an in-depth investigation of news typology to transfer the DSF to specific types of news samples at a fine-grained level. Secondly, while LIMFA proves to be effective, interpreting the extracted features is challenging. Based on previous studies, DIF in fake news detection are the common trait or attribute shared by many types of fake news, regardless of the domain. These could include features related to the use of sensational language, writing style, the presence of conspiracy theories, and patterns of stances. As for DSF, they can be attributed to the aspects of news articles that are tied to the source or domain from which the news originates. These could include the use of specific vocabulary, the publication date, or the presence of particular keywords that are common within a specific domain. However, the explanation of universal features of fake news detection lacks consensus and empirical studies. We encourage further efforts to find the universal knowledge of new samples and subsequently guide representation learning based on this knowledge.

Appendix A: Overall results of Table 12

Overall results using LOO-CV on the CH-9 dataset. See Table 12.

Table 12 Overall results of cross-domain models using leave-one-domain-out cross-evaluation on the CH-9 dataset

	Method	Fina.		Edu.		Mil.		Sci.		Soc.	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G2	EDDFN	0.8762	0.8452	0.8566	0.8562	**0.8744**	**0.8617**	0.8483	0.8405	0.8193	0.8158
	MDFEND	0.8867	0.8442	0.8415	0.8441	0.8328	0.8191	0.8527	0.8387	0.8329	0.8326
G3	EANN	0.8919	0.862	0.8592	0.8585	0.8455	0.8232	0.8639	0.84	0.8556	0.8553
	BDANN	0.9043	0.8498	0.8621	0.8563	0.8485	0.8384	0.8593	0.8354	0.8634	0.8623
	Ours	**0.9188***	**0.8732***	**0.8634***	**0.8676***	0.8580	0.8424	**0.8705***	**0.8424***	**0.8646**	**0.8695**

	Method	Ent.		Heal.		Disa.		Poli.		Overall	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
G2	EDDFN	0.8776	0.8531	0.8572	0.8563	0.8759	0.8434	0.8700	0.8591	0.8479	0.8639
	MDFEND	0.8913	0.8600	0.8296	0.8294	0.8402	0.8091	0.8506	0.8377	0.8509	0.8350
G3	EANN	0.8754	0.8499	0.8417	0.8541	0.8676	0.8364	0.859	0.8536	0.8622	0.8481
	BDANN	0.8889	0.8607	0.8561	0.8586	**0.8881**	**0.8587**	0.8693	0.8614	0.8711	0.8535
	Ours	**0.8925**	**0.8683**	**0.8742***	**0.8741***	0.8764	0.8439	**0.8797***	**0.8710***	**0.8778**	**0.8560**

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Data availability All data, models and code generated or used during this study are available at https://github.com/TAN-OpenLab.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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